

Jan 19, 2024

Introduction to Computer and Network Security

Part 1

Question 1 [50 points]

In the context of TLS and the PKI, describe:

- (a) what a digital certificate is and what are its main components [10 points];
- (b) what a root CA is [5 points];
- (c) how a system can validate a certificate [5 points];
- (d) how TLS 1.2 handshake works by drawing the message sequence chart and briefly describing each step [25 points];
- (e) how TLS ensures authentication, confidentiality, and integrity [5 points].

Part 2

Question 2 [40 points]

In the context of password based authentication, describe (a) Multi-factor authentication, (b) what is its purpose, (c) the three available factors, along with an example and (d) the three NIST-defined assurance levels.

Question 3 [10 points]

Describe the technique of k-anonymity and explain how it can mitigate linkage attacks.

Please, remember...

Be sure to annotate the following information on each used sheet of paper (possibly, 2 sheets at most)

- Name
- Surname
- Student ID number
- Question number

